OpenAI

no java before code

Bro, this line:

```Python
vectorizer = TfidfVectorizer(stop_words='english')
```

does **NOT** convert text into numbers yet.

It only creates a TF-IDF Vectorizer object and stores it in:

```
vectorizer
```

Think of it like:

```
model = MultinomialNB(...)
```

At that moment:

```
model
```

is not trained yet.

Similarly:

```
vectorizer = TfidfVectorizer(...)
```

creates the vectorizer, but it hasn't processed any text yet.

What is TfidfVectorizer?

It's a class from Scikit-Learn.

Its job is:

```
Text
↓
Numbers
```

Machine learning models understand numbers.

They do NOT understand:

```
"free money now"
```

or

```
"hello friend"
```

directly.

So we need something that converts words into numbers.

That's what TF-IDF does.

What does stop_words='english' mean?

```
TfidfVectorizer(stop_words='english')
```

means:

```
Ignore common English words.
```

Examples:

```
the
is
am
are
was
were
and
of
to
in
```

These words appear in almost every message.

So they usually don't help determine:

```
SPAM
or
HAM
```

Example

Message:

```
the free money is waiting for you
```

Without stop-word removal:

```
the
free
money
is
waiting
for
you
```

With stop-word removal:

```
free
money
waiting
```

Important words remain.

Why remove them?

Suppose every message contains:

```
the
is
and
to
```

These words don't help identify spam.

But words like:

```
free
winner
cash
prize
lottery
offer
```

are much more useful.

So TF-IDF focuses on important words.

Then when does conversion happen?

Not here:

```
vectorizer = TfidfVectorizer(...)
```

Conversion happens later:

```
X_vectorized = vectorizer.fit_transform(X)
```

This is where TF-IDF actually reads messages and creates numerical vectors.

Think of it like this

```
vectorizer = TfidfVectorizer(...)
```

↓

```
Create the machine
```

Then:

```
vectorizer.fit_transform(X)
```

↓

```
Put messages into the machine
```

↓

```
Get numerical vectors out
```

Notes Format

```
TfidfVectorizer → sklearn class

vectorizer → variable storing the TF-IDF object

stop_words='english' → removes common English words

Examples:
the, is, am, are, and, of, to

These words usually provide little value for spam detection.

👉 Use: create a TF-IDF vectorizer that will later convert text into numerical features while
ignoring common English stop words.
```